

MCP Dev Summit Toronto 2026

Multi-Agent Systems in Practice:

What Our Experiments Taught Us

OCTOBER 6, 2026

AI Engineering Is Balancing Cost, Speed and Capability



Cost

Use fewer tokens and lighter-weight models.



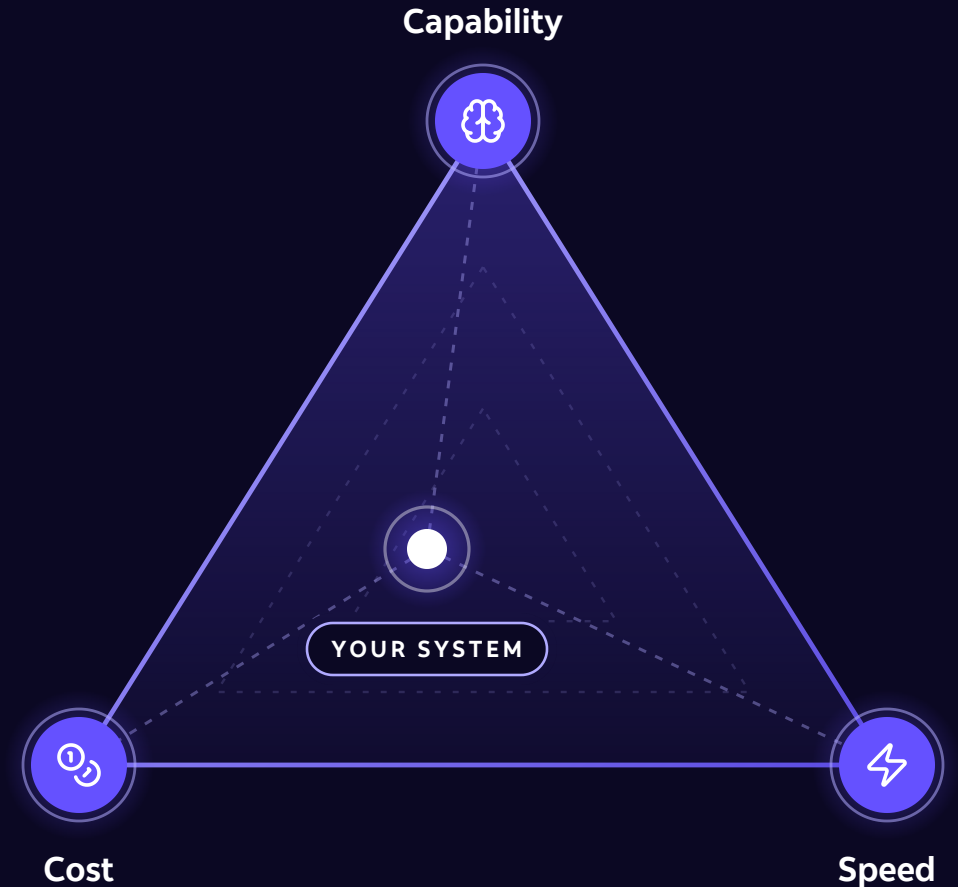
Speed

Run work in parallel, and cache what repeats.



Capability

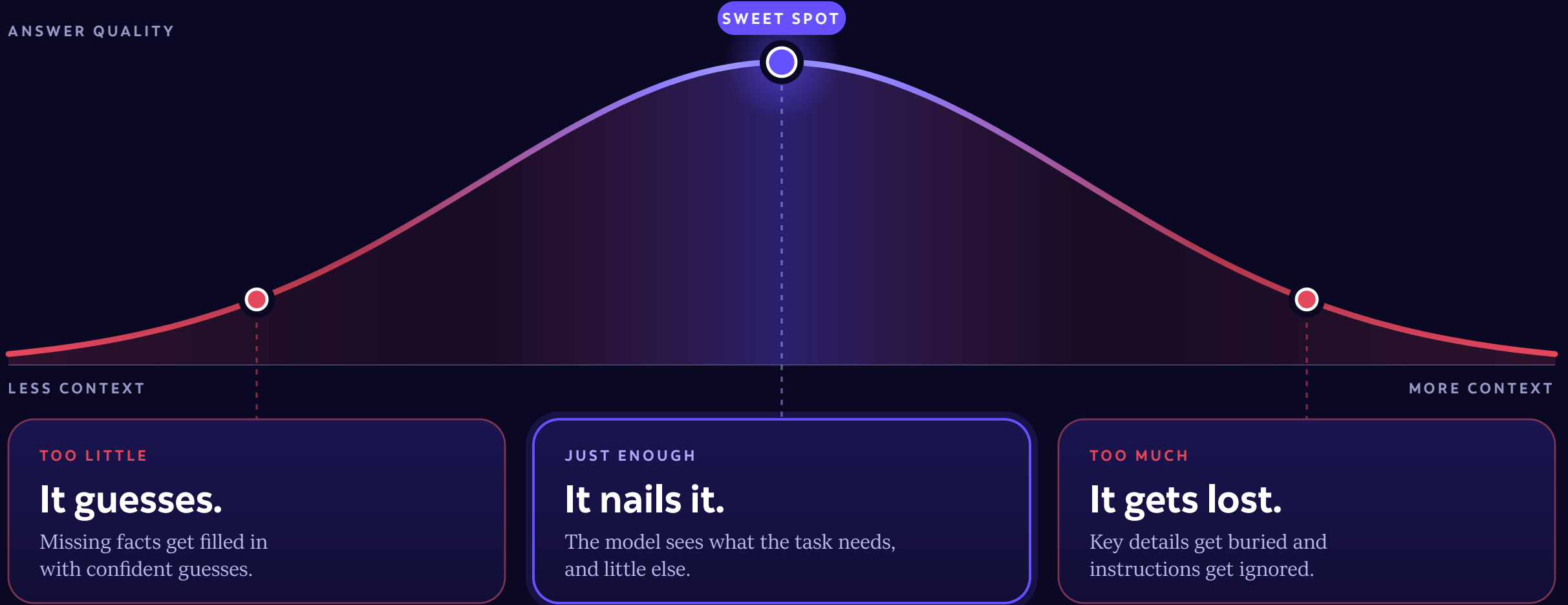
A stronger model or more reasoning helps, but mostly: give each call the right data context.



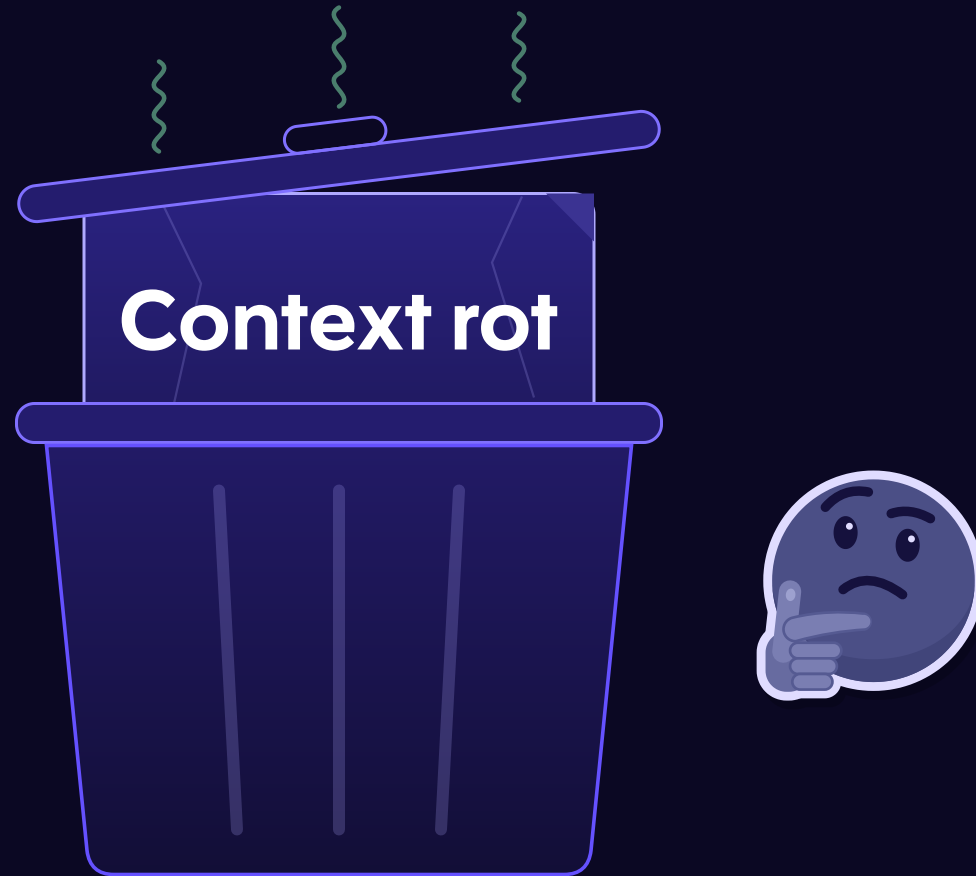
Investigating: Capability Is a Context Problem

What goes in shapes what comes out.

ANSWER QUALITY



Agentic Patterns For:

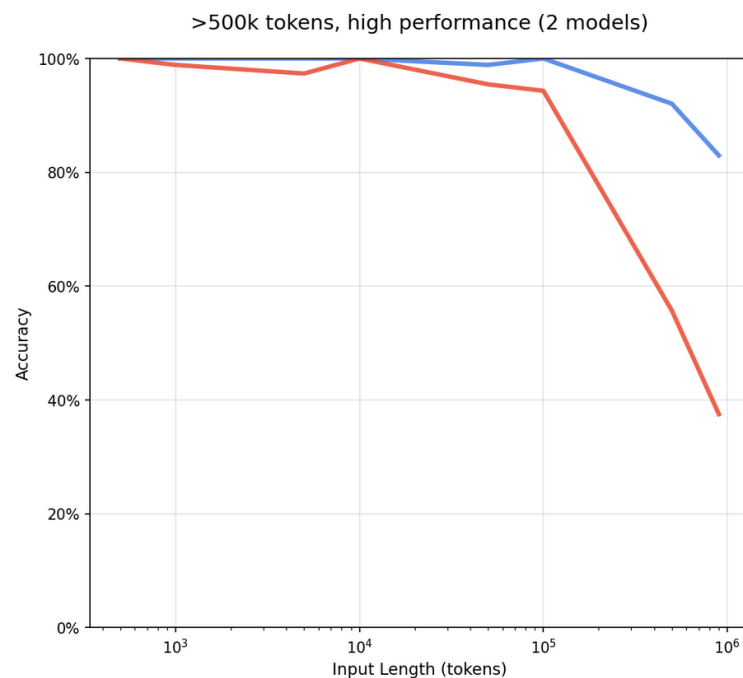


Context Rot, Measured

Chroma popularized the term in July 2025: models get worse as the input grows, even on simple tasks.

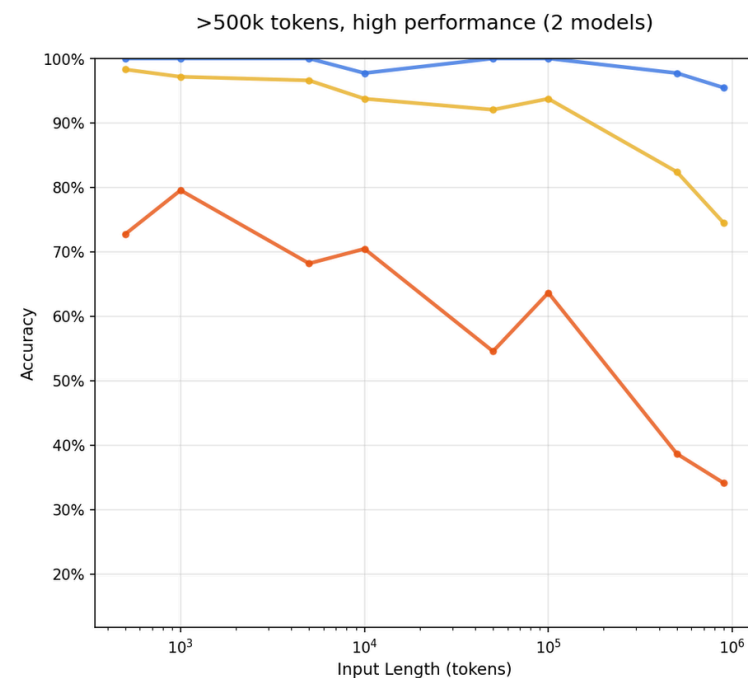
NEEDLE THAT DOESN'T ECHO THE QUESTION

— High similarity — Low similarity



LOOK-ALIKE DISTRACTORS

— No distractors — One — Four



Models tested: 18, including Claude Opus 4, o3, GPT-4.1, Gemini 2.5 Pro and Qwen3-235B. Both panels show the two best models with windows over 500K tokens; Chroma doesn't name which two.

Source: Chroma Research, "Context Rot: How Increasing Input Tokens Impacts LLM Performance" (Hong, Troynikov, Huber), July 14, 2025.

Targeting the Limiting Factor

Long context used to be the limiting factor. The people testing models say it is being solved.

“

For text, Gemini 1.5 Pro achieves 100% recall up to 530k tokens, 99.7% up to 1M tokens, and 99.2% accuracy up to 10M tokens.

JD

Jeff Dean @JeffDean
Then Chief Scientist, Google

Gemini 1.5 Pro · Feb 2024

“

Long context improvements are a HUGE deal. This unlocks 1mil context windows to actually work. Now GPT doesn't randomly forget 9-30% of your context.

ND

Nick Dobos @NickADobos
Prompt engineer, The Browser Company

GPT-6 Astra · Sep 2026

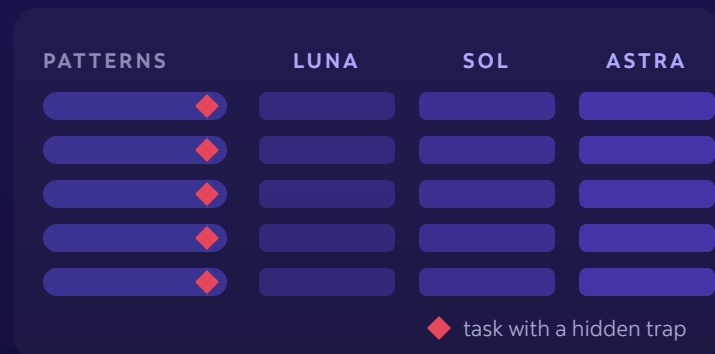
If the window holds, the limiting factor moves to cost and speed.

Context Evals

Our own eval set, built to test agentic patterns and context strategies, not just models.

01 · HOW WE TEST

Patterns and Context, Not Just Models



02 · FIRST RUN

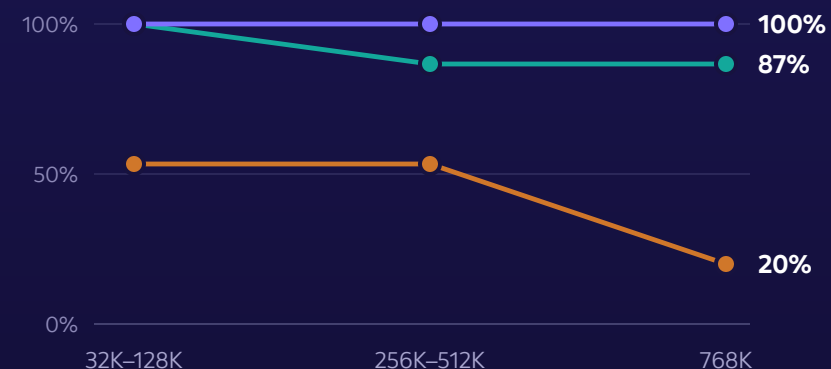
100%

Sol and Astra on
needle-in-a-haystack
questions



03 · HARDER SET: RESULTS

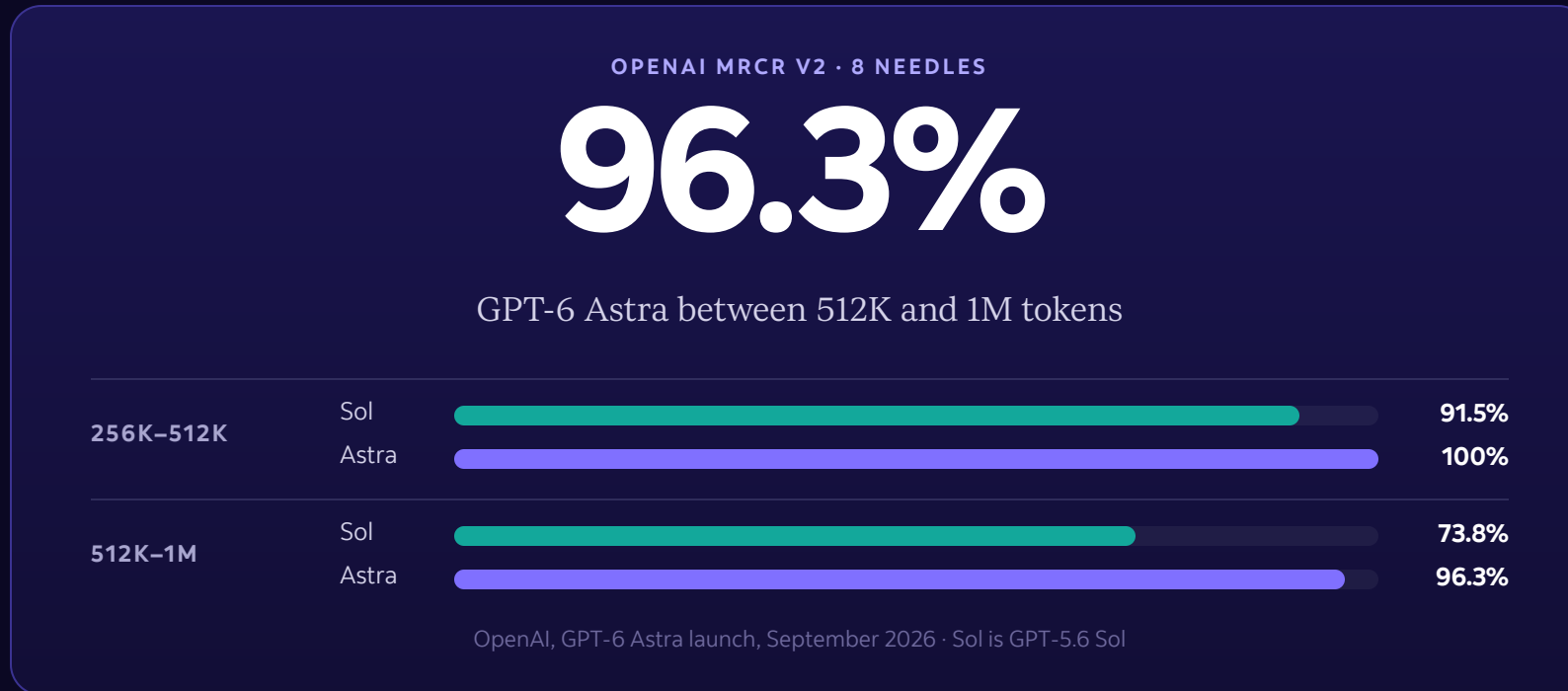
- Astra** Perfect to 768K on the two hardest types
- Sol** Perfect to 128K; from 256K misses long counting
- Luna** Misses even at 32K; 20% right at 768K



The test: one decision at the end of a replayed support shift. The rule is stated once at the top; the facts it needs hide among hundreds of look-alike cases. One shot, no search tool.

Context Windows Are Becoming Fully Searchable

In 2026, frontier models keep their accuracy across the whole window.



For frontier models, finding what is in the window is no longer the hard part.

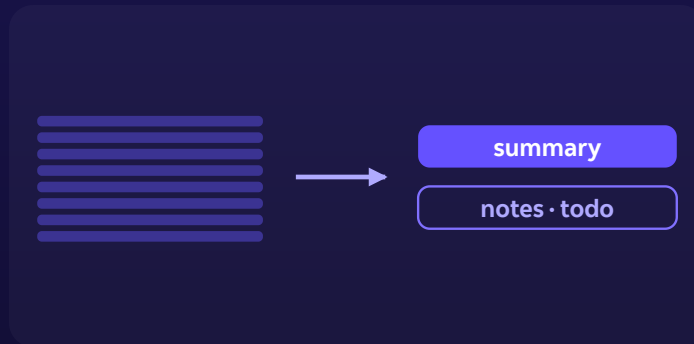
Three Ways To Manage Context

Context strategies keep one agent and manage what it sees. Agentic patterns split the work across agents.

CONTEXT STRATEGY

Compress it

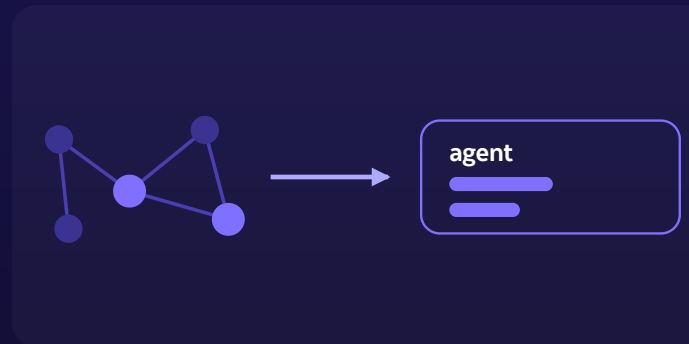
Keep one agent; shrink its history to fit the window.



CONTEXT STRATEGY

Fetch it

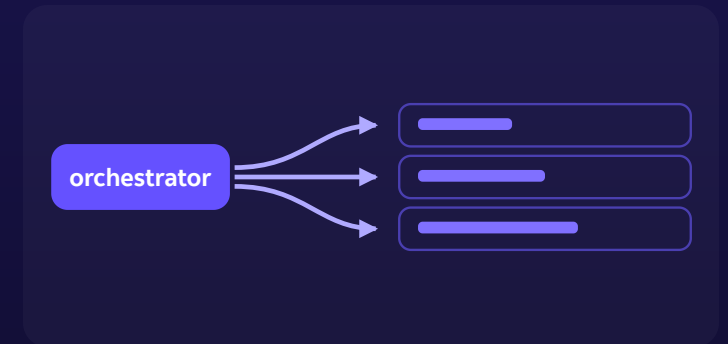
Keep one agent; pull in only what the step needs.



AGENTIC PATTERNS

Split it

Spread work across agents, each with its own slice.

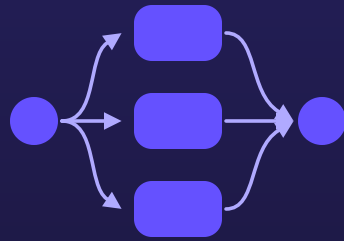


Five Multi-Agent Orchestration Patterns

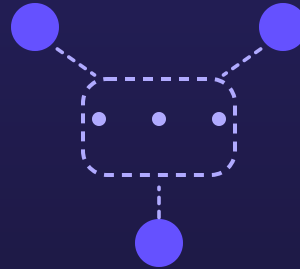
Same agents, five different ways to coordinate them.



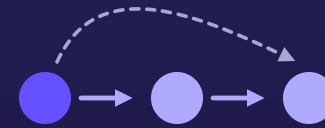
Sequential



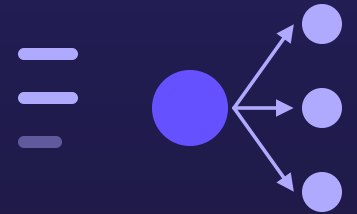
Concurrent



Group Chat



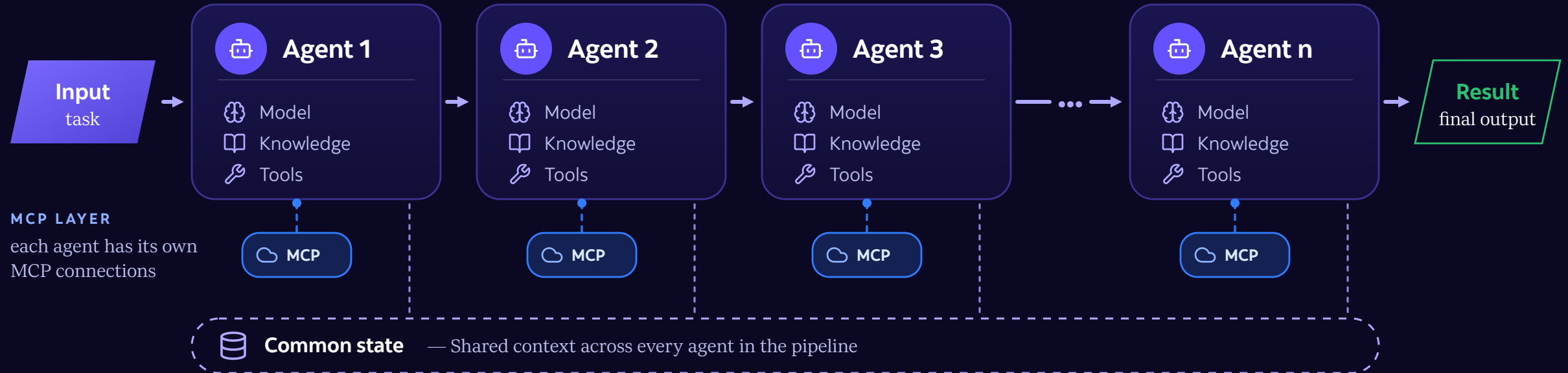
Handoff



Magentic

Sequential Orchestration

Agents are chained in a predefined order. Each one processes the previous agent's output.



1 Input enters Agent 1

It reasons with its own model, knowledge and tools.

2 Output becomes the next input

Each agent builds on the previous one, down to Agent n.

3 The last output is the result

Order is fixed up front: routing is deterministic.

BEST FOR

Clear linear dependencies and step-by-step refinement.

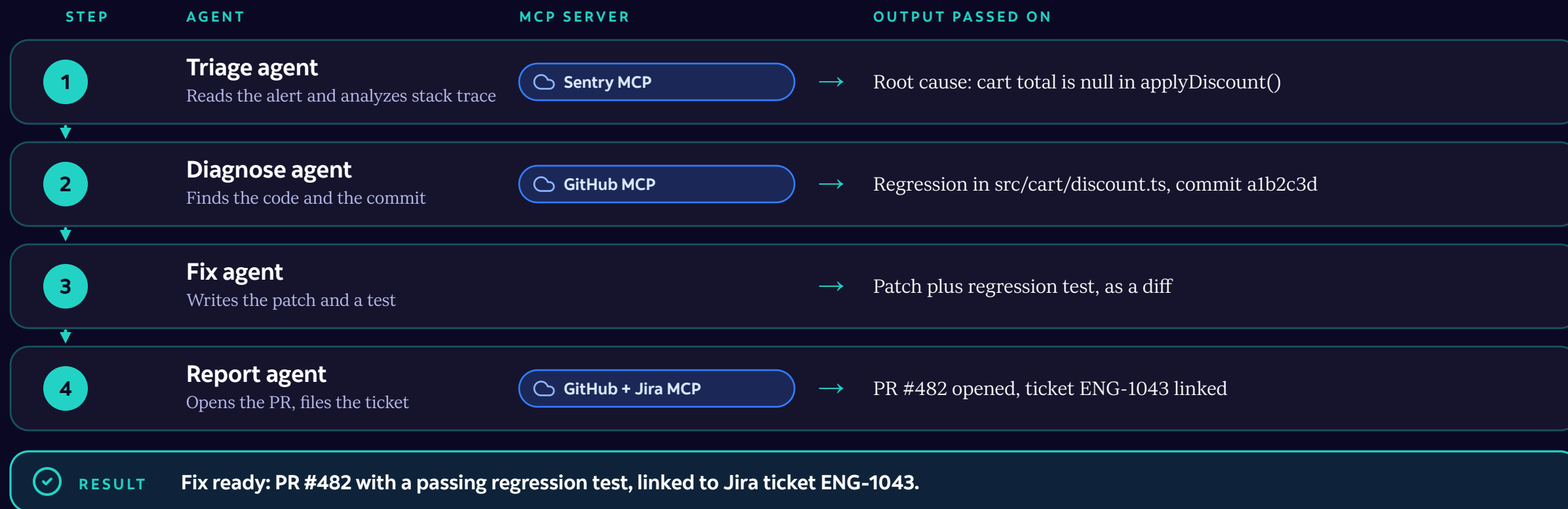
WATCH OUT FOR

Early errors flow downstream, and there is no parallelism.

Bug Triage to Pull Request

WORKED EXAMPLE

Trigger: an e-commerce store's checkout fails with HTTP 500 errors after release 2.14.1, and Sentry raises an alert.



WHY SEQUENTIAL FITS

Each step needs the previous step's output, so the order is fixed.

MCP IN THIS FLOW

Only agents that need outside data connect to MCP servers.

Concurrent Orchestration



Agents work on the same input at the same time. A collector combines their results.



BEST FOR

Independent perspectives and time-sensitive work.

WATCH OUT FOR

Conflicting results, cost multiplied by N, and shared-state races.

Parallel Pull Request Review

Trigger: a pull request is opened and needs a review from several angles.



WHY CONCURRENT FITS

The four reviews are independent, so total time is the slowest one.

MCP IN THIS FLOW

Only reviewers that need outside data connect; the others read the diff.

Group Chat Orchestration

Agents discuss in one shared thread. A chat manager decides who speaks and when to stop.



BEST FOR

Consensus building, brainstorming, and maker-checker review.

WATCH OUT FOR

Endless loops, growing context, and control past three agents.

Database Migration Review

WORKED EXAMPLE

Trigger: a schema migration needs agreement before it goes to production.



Chat manager

orchestrates only

Picks who speaks next
Stops once the reviewer
approves the plan

HOW IT RUNS

Every agent reads the whole thread,
then adds its turn. Tool results
are trimmed before they are posted.



Shared thread: review of migration 0042_add_status.sql

1

Author agent

Proposes: add status column with constraint NOT NULL, and add an index

2

Schema reviewer

Adding the NOT NULL constraint scans the whole table under a lock that blocks queries



Postgres MCP

3

Author agent

Revises: add a nullable status column, backfill in batches then add the constraint

4

Performance agent

EXPLAIN confirms the status query uses the new index



Postgres MCP

5

Chat manager

Reviewer approves the revised plan. Consensus reached, stop.



RESULT

Approved migration plan: nullable column, batched backfill, then the constraint.

WHY GROUP CHAT FITS

A maker and checkers need to challenge each other until they agree.

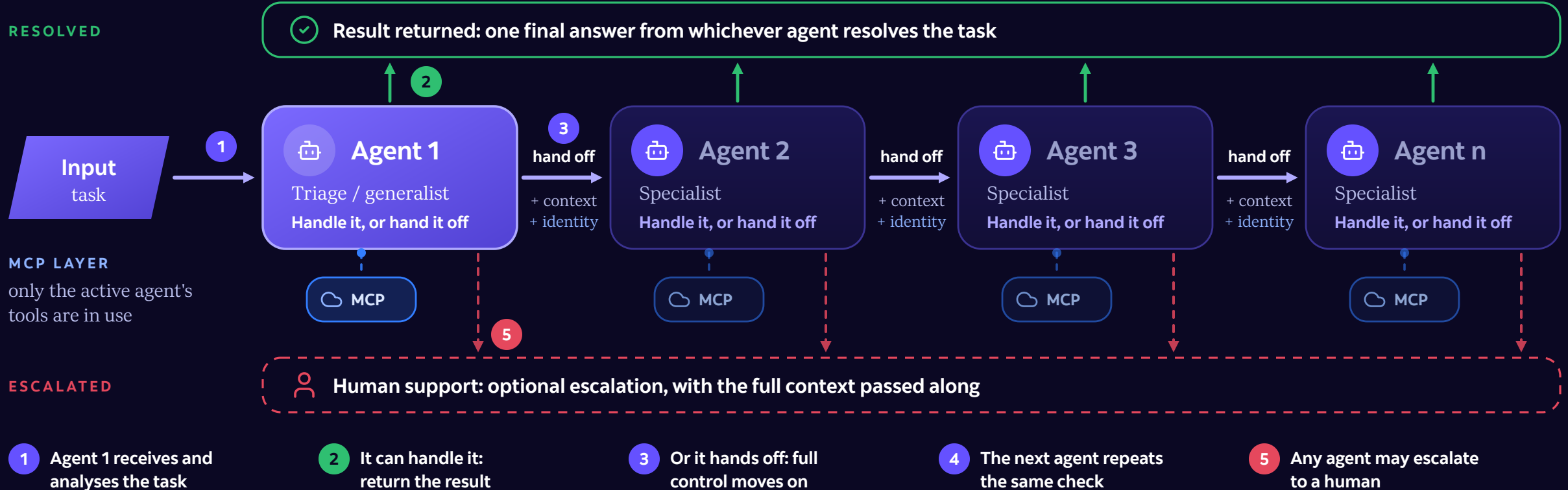
MCP IN THIS FLOW

Only the database reviewers connect to a Postgres MCP server.

Handoff Orchestration



Agents pass full control to one another until the right one resolves the task.



BEST FOR

The right specialist only becomes clear during processing.

WATCH OUT FOR

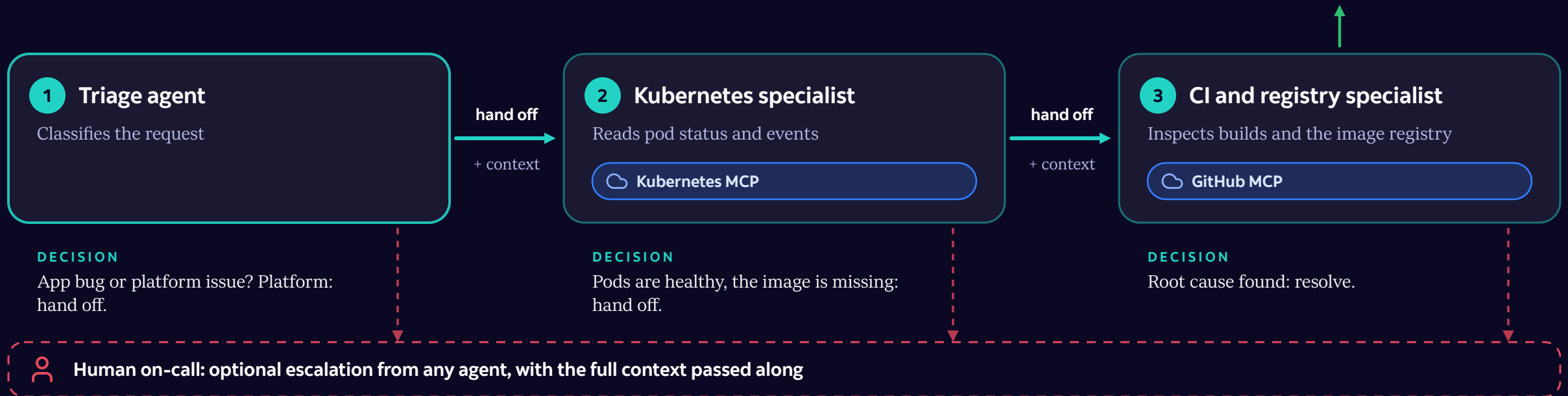
Infinite handoff loops and unpredictable routing paths.

Deployment Failure Support

WORKED EXAMPLE

Trigger: a developer reports that orders-service pods are stuck in ImagePullBackOff.

✓ Resolved: the last agent returns one answer, with the pipeline re-run and the image published



WHY HANDOFF FITS

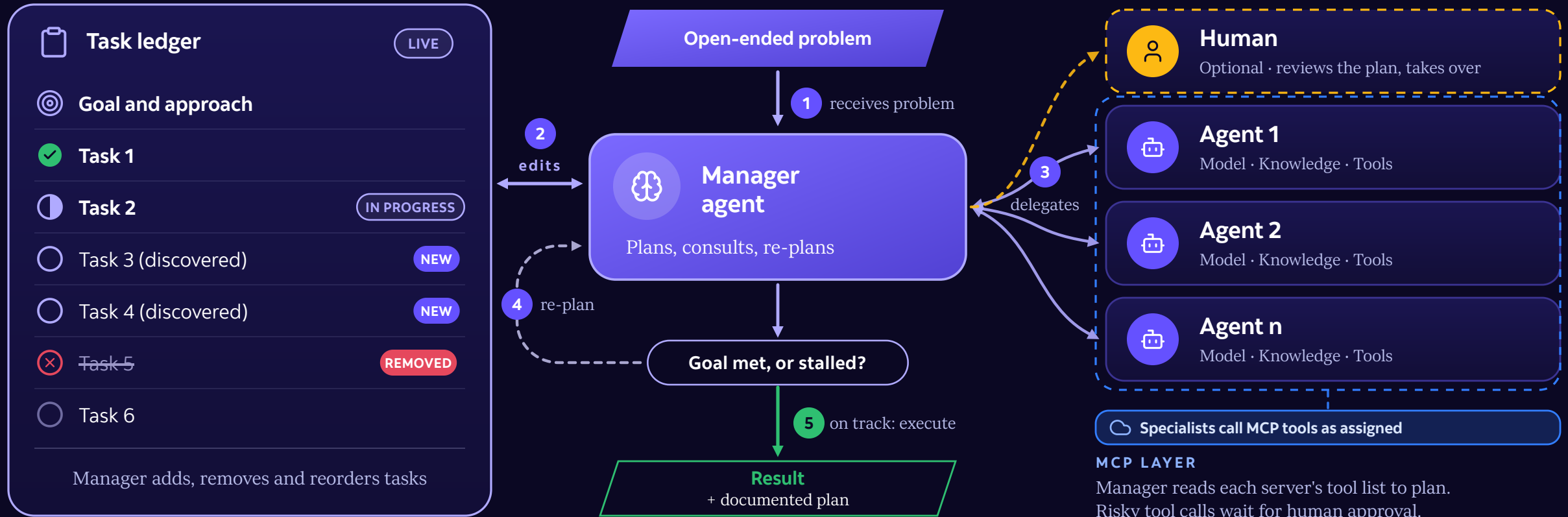
The right specialist only becomes clear as the problem is investigated.

MCP IN THIS FLOW

Only the active agent's tools are in use; the triage agent needs none.

Magentic Orchestration

A manager builds and revises a task ledger with specialist agents, then executes the plan.



BEST FOR

Open-ended problems with no predetermined solution path.

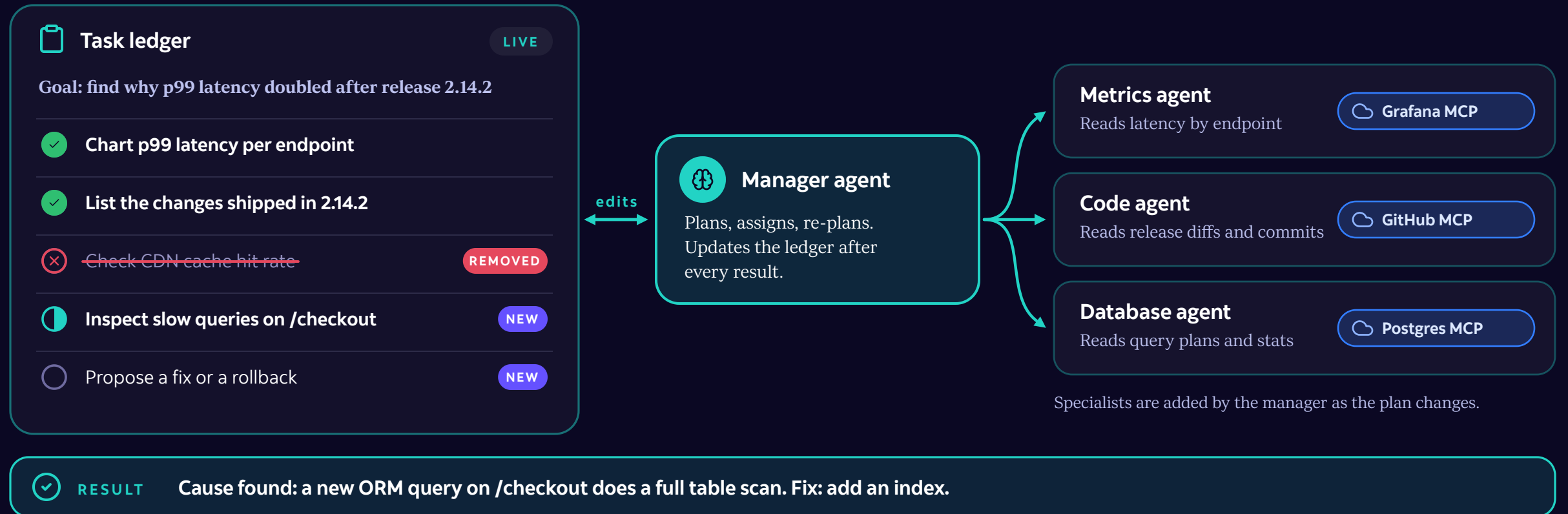
WATCH OUT FOR

Slow convergence, stalls on vague goals, and high token cost.

Latency Regression Investigation

WORKED EXAMPLE

Trigger: p99 latency doubled after release 2.14.2 and nobody knows why.



WHY MAGENTIC FITS

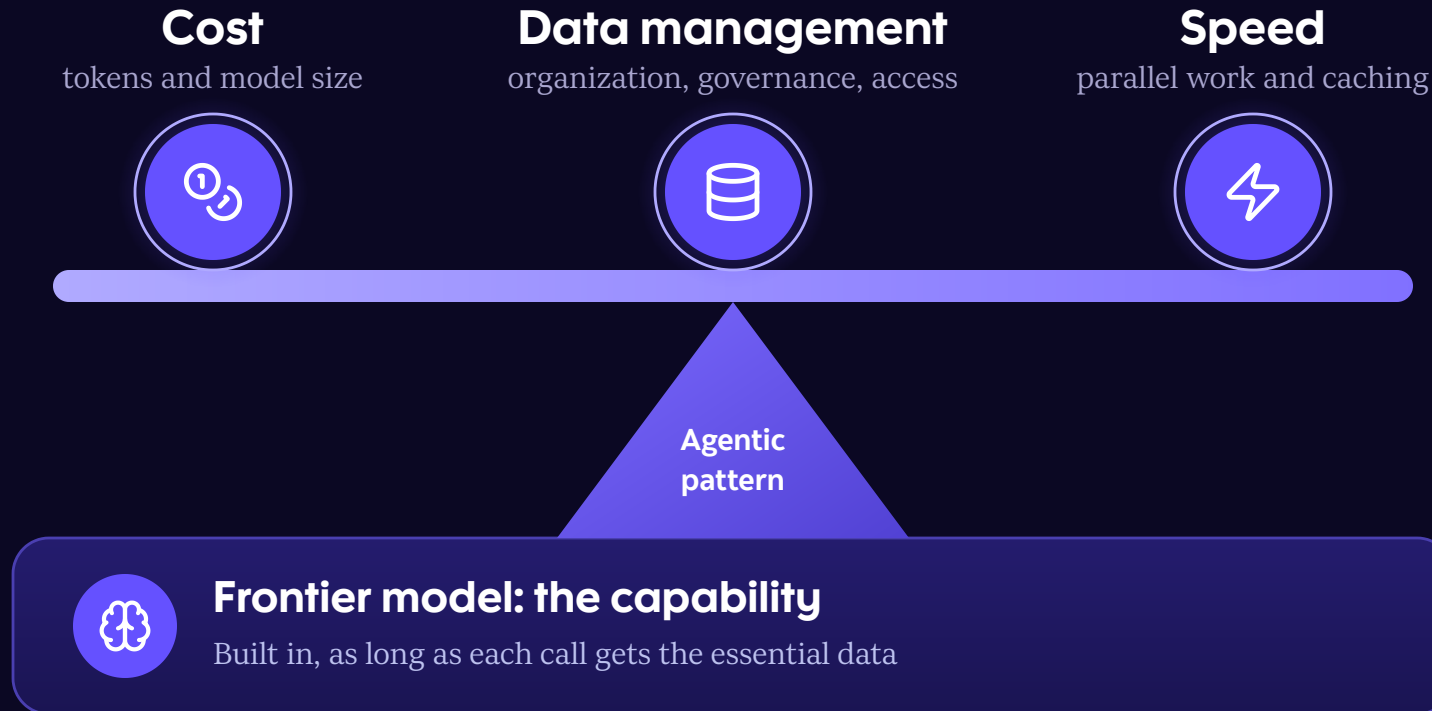
The steps are unknown upfront: each finding changes the plan.

MCP IN THIS FLOW

Each specialist connects to the MCP server for its own domain.

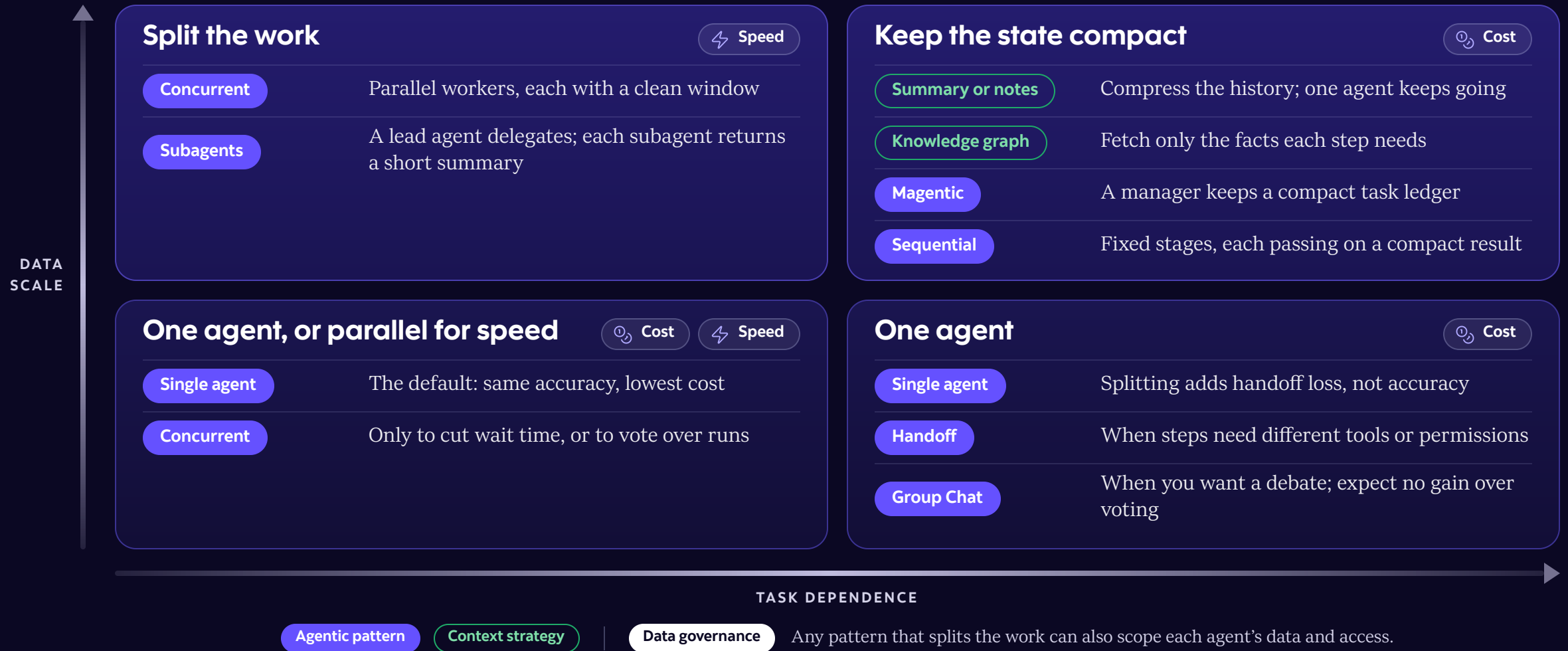
Agentic Systems into 2027

The model brings the capability. The agentic pattern helps you manage speed, cost and data organization.



Choosing a Pattern

Two questions: does everything fit in the context window, and does each step depend on the last?



MCP DEV SUMMIT · TORONTO 2026

Thank you.

Questions? Come find us after the session, or
pick up the code.



Get the code

[github.com/Erik-Koning/
agentic-patterns-eval](https://github.com/Erik-Koning/agentic-patterns-eval)

The five patterns, the context strategies
and the full eval set.

TEAM



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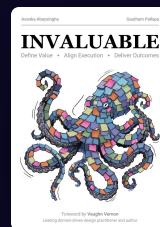
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Invaluable and Mastering Enterprise Platform Engineering,
co-authored by Gautham Pallapa

Appendix: When to Use Each Pattern

Sequential

USE WHEN

Fixed stages, each building on the last.

CONTEXT

Each agent sees only the previous output.

WATCH OUT FOR

Details lost between stages; the slowest option.

OUR EVIDENCE

Partly tested

Workers in series match parallel accuracy but take about twice as long.

Concurrent

USE WHEN

Independent subtasks, or several takes on one input, when speed matters.

CONTEXT

Each worker starts with a clean window.

WATCH OUT FOR

2–3× the tokens; results to merge.

OUR EVIDENCE

Tested

About 20 points over one agent on 32 suppliers. Voting matches debate at equal cost.

Group Chat

USE WHEN

You want a debate or review between perspectives.

CONTEXT

One shared thread that grows every turn.

WATCH OUT FOR

About 5× the cost; slow to converge.

OUR EVIDENCE

Tested

A debating council is expected to do no better than voting at the same cost.

Handoff

USE WHEN

Steps need different skills, tools or permissions.

CONTEXT

The next agent gets only what is handed over.

WATCH OUT FOR

Handoffs drop context; routing loops.

OUR EVIDENCE

Partly tested

Specialist roles expected to add nothing beyond isolation. Permissions are judgement.

Magentic

USE WHEN

Open-ended work where the plan emerges, or approvals need a record.

CONTEXT

A manager's task ledger holds the plan; specialists get focused tasks.

WATCH OUT FOR

Loops and unpredictable cost; the manager's context grows.

OUR EVIDENCE

Not tested

From practice.

EXPECTED

Specialise for permissions, ownership and maintainability, not for accuracy.